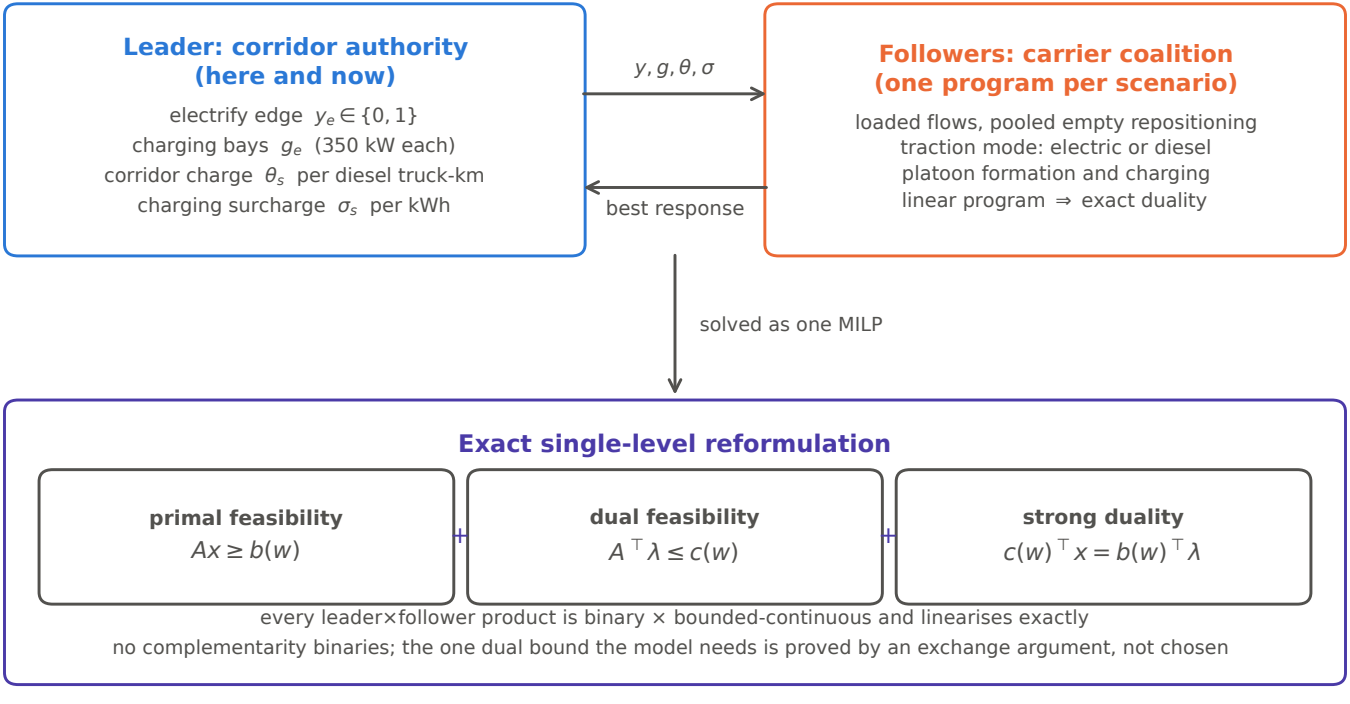
Leader: corridor authority
(here and now)
electrify edge $y_e \in \{0,1\}$
charging bays g_e (350 kW each)
corridor charge θ_s per diesel truck-km
charging surcharge σ_s per kWh
y, g, θ, σ
best response
Followers: carrier coalition
(one program per scenario)
loaded flows, pooled empty repositioning
traction mode: electric or diesel
platoon formation and charging
linear program ⇒ exact duality
solved as one MILP
Exact single-level reformulation
primal feasibility
$Ax \geq b(w)$
+
dual feasibility
$A^\top \lambda \leq c(w)$
+
strong duality
$c(w)^\top x = b(w)^\top \lambda$
every leader×follower product is binary × bounded-continuous and linearises exactly
no complementarity binaries; the one dual bound the model needs is proved by an exchange argument, not chosen